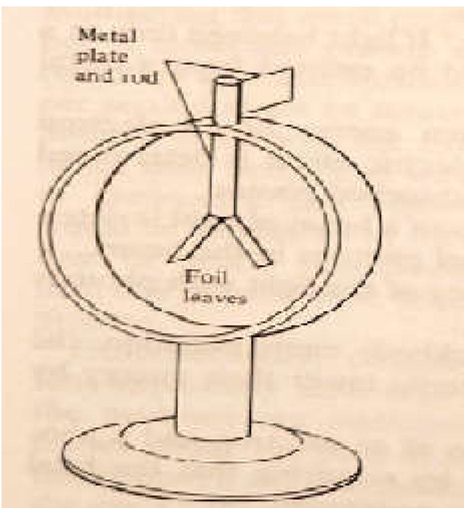
**26.1]** When light of frequency $7\times10^{14}$ Hz shines on a metal surface, electrons with a maximum speed of $6\times10^{5}$ m/s are emitted. What is the photoelectric threshold frequency of the sample?

**26.2]** The work function of metallic sodium is 2.3ev. What is the maximum wavelength of the incident light for which photoemission of electros will occur?

**26.5]** An electroscope has two metal foil leaves that normally hang side by side. They are both connected to a metal plate (Fig.26.10). If the light of a sufficiently high frequency is shone on the plate, the leaves are observed to spread apart. Describe what causes this.




**26.10]** In a monochromatic beam of light of wavelength 500nm, what is the photon (a)energy, (b) momentum

26.14] In Compton scattering experiment, the initial and final photon frequencies are2.4×10^20Hz and 1.6×10^20Hz, respectively. What is the recoil energy of thescattering electron in electron volt?

26.17] The electrons in an X-ray tube are accelerated from rest through a potential difference of 50000V. (a) What is the maximum frequency X-ray photon is produced. (b)What is the wavelength of this photon?